



Web Development Technologies

Tutorial and Lab Sheet 4 Solution Guide

Exercise 1 Code First approach

Ask your tutor about the following:

- a. What is difference between Code First and Database First approaches?

Code First: Database tables do not exist.

You have full control over the code rather than database activity. In this approach, we can do all the database operations from the code and manual changes to database have been lost and everything is depending on the code. In this you need to create entities as data model.

Database First: Database tables and constraints preexist. You have to reverse engineer Model and context classes from those in your ASP.NET Core project. We will cover D/B first approach during lecture 7.

Each approach has its pros and cons.

Code First

Advantages

- You can create the database and tables from your model entities
- You can specify which related collections are to be eager loaded, or not be serialized at all.
- Full control over database version control.
- Good for small applications.

Disadvantages

- You have to write everything related to database in the visual studio code.
- If you want to change anything in the database tables you to make changes in the entity classes in the code file and run the update-database from the package manager console.
- Not preferred for Data intensive applications.



Database First

Advantages

- Mapping and creation of keys and relationships are easy as you need not have to write any code
- Preferred for data intense and large applications

Disadvantages

- Need to know SQL and database constraints

b. What is Entity Framework (EF)?

Entity Framework Core is a modern object-database mapper for .NET. It supports LINQ queries, change tracking, updates, and schema migrations. EF Core works with many databases, including SQL Database (on-premises and Azure), SQLite, MySQL, PostgreSQL, and Azure Cosmos DB.

The idea is to write database operations in C# and no SQL needs to be written, it is auto generated behind the scenes.

c. What is LINQ to Entities?

Please read:

<https://learn.microsoft.com/en-au/dotnet/framework/data/adonet/ef/language-reference/linq-to-entities>

As for example: <https://learn.microsoft.com/en-au/aspnet/core/tutorials/first-mvc-app/start-mvc?tabs=visual-studio&view=aspnetcore-10.0>

You will have to make following changes in above tutorial:

Part 4 - Add a model to an ASP.NET Core MVC app

Add a database connection string

The connection string in the tutorial won't work for you:

```
"ConnectionStrings": {  
  "MvcMovieContext": "Server=(localdb)\\mssqllocaldb;Database=MvcMovieContext-1;Trusted_Connection=True;MultipleActiveResultSets=true"  
}
```

Change it to:

```
"ConnectionStrings": {  
  "MvcMovieContext": "Server=rmit.australiaeast.cloudapp.azure.com;  
Encrypt=False;Uid=<YOUR_USERNAME>;Pwd=<YOUR_PASSWORD>;Database=<YOUR_DATABASE>"  
}
```

Replace <YOUR_USERNAME> with the username you have been given such as s1234567_a2 where 1234567 is your student number; similar change <YOUR_PASSWORD> and <YOUR_DATABASE>.

If you still don't know what is being talked about here and you did not attend tutorial session:

- Watch recording of Lecture 4 or
- Look at appsettings.json file in solution archive for tutorial this week

Part 5 - Work with a database in an ASP.NET Core MVC app

Ensure you use SQL Server and not Sqlite for the database. The following should be present within Program.cs to configure the database:

```
builder.Services.AddDbContext<MvcMovieContext>(options =>  
    options.UseSqlServer(builder.Configuration.GetConnectionString("MvcMovieContext")));
```

Once again if you don't know what is being talked about here and you did not attend tutorial session

- Watch recording of lecture 4 or
- Look at Program.cs file in solution archive for tutorial this week

Exercise 2 and 3

These do not require any changes following MSDN tutorial verbatim.

***NOTE:** In an advanced elective like this, you will not find line by line explanation of each code in solution files. You are expected to attend tutorial / lab sessions, ask questions in class or on the Canvas Discussions forum, watch recordings, etc...*